OpenDRIVE® 1.8.1, Static Road Network Description, 2024-11-21

Annex A: Enumerations

A.1 Core

A.1.1 e_unitDistance

Table 149. Enumerations of the unitDistance attribute

Name	Type
`ft`	string
`km`	string
`m`	string
`mile`	string

A.1.2 e_unitSlope

Table 150. Enumerations of the unitSlope attribute

Name	Type
`%`	string

A.1.3 e_dataQuality_RawData_Source

Table 151. Enumerations of the dataQuality RawData Source attribute

Name	Type
`cadaster`	string
`custom`	string
`sensor`	string

A.1.4 e_unitSpeed

Table 152. Enumerations of the unitSpeed attribute

Name	Type
`km/h`	string
`m/s`	string
`mph`	string

A.1.5 e_dataQuality_RawData_PostProcessing

Table 153. Enumerations of the dataQuality RawData PostProcessing attribute

Name	Type
`cleaned`	string
`fused`	string
`processed`	string
`raw`	string

A.1.6 e_unitMass

Table 154. Enumerations of the unitMass attribute

Name	Type
`kg`	string
`t`	string

A.1.7 t_yesNo

Table 155. Enumerations of the yesNo attribute

Name	Type
`no`	string
`yes`	string

A.2 Junction

A.2.1 e_junction_type

Table 156. Enumerations of the junction type attribute

Name	Type	Introduced	Description
`crossing`	string	1.8.0	Unconnected crossing, e.g. railroad or pedestrian
`default`	string		Common junction, used if no value is specified
`direct`	string	1.7.0	Direct junction, e.g. entries and exits

Name	Type	Introduced	Description
`virtual`	string		Virtual junction, e.g. driveways and entries to parking lots

A.2.2 e_road_surface_CRG_purpose

Table 157. Enumerations of the road surface CRG purpose attribute

Name	Type
`elevation`	string
`friction`	string

A.2.3 e_road_surface_CRG_mode

Table 158. Enumerations of the road surface CRG mode attribute

Name	Type	Description
`attached0`	string	ASAM OpenCRG reference line: is discarded. Total height: OpenCRG height only
`attached`	string	ASAM OpenCRG reference line: is discarded. Total height: ASAM OpenDRIVE height + OpenCRG height
`genuine`	string	ASAM OpenCRG reference line: shifted and rotated so beginning of reference line matches position given in ASAM OpenDRIVE. Total height: OpenCRG height only
`global`	string	ASAM OpenCRG reference line: taken unmodified. Total height: OpenCRG height only

A.2.4 e_junction_segment_type

Table 159. Enumerations of the junction segment type attribute

Name	Type	Introduced
`joint`	string	1.8.0
`lane`	string	1.8.0

A.2.5 e_junctionGroup_type

Table 160. Enumerations of the junctionGroup type attribute

Name	Type	Introduced	Description
`complexJunction`		1.8.0	for example junctions with slip lanes
`highwayInterchange`		1.8.0	for example clover leaf interchange
`roundabout`	string		
`unknown`	string		

A.2.6 e_elementDir

Table 161. Enumerations of the elementDir attribute

Name	Type
`+`	string
`-`	string

A.2.7 e_contactPoint

Table 162. Enumerations of the contactPoint attribute

Name	Type
`end`	string
`start`	string

A.2.8 e_connection_type

Table 163. Enumerations of the connection type attribute

Name	Type
`default`	string
`virtual`	string

A.3 Lane

A.3.1 e_roadMarkWeight

Table 164. Enumerations of the roadMarkWeight attribute

Name	Type
`bold`	string
`standard`	string

A.3.2 e_laneAdvisory

Table 165. Enumerations of the laneAdvisory attribute

Name	Introduced	Description
`both`	1.8.0	May be used by traffic from both adjacent lanes as an advisory lane.
`inner`	1.8.0	May be used by traffic from the inner lane as an advisory lane.
`none`	1.8.0	May not be used as an advisory lane.
`outer`	1.8.0	May be used by traffic from the outer lane as an advisory lane.

A.3.3 t_bool

Table 166. Enumerations of the bool attribute

Name	Type
`false`	string
`true`	string

A.3.4 e_roadMarkType

Table 167. Enumerations of the roadMarkType attribute

Name	Type	Description
`botts dots`	string	
`broken broken`	string	From inside to outside, exception: center lane – from left to right
`broken solid`	string	From inside to outside, exception: center lane – from left to right
`broken`	string	
`curb`	string	

Name	Type	Description
`custom`	string	If detailed description is given in child tags
`edge`	string	Describing the limit of usable space on a road
`grass`	string	Meaning a grass edge
`none`	string	
`solid broken`	string	From inside to outside, exception: center lane – from left to right
`solid solid`	string	For double solid line
`solid`	string	

A.3.5 e_accessRestrictionType

Table 168. Enumerations of the accessRestrictionType attribute

Name	Type	Introduced
`HOV`	string	1.8.0
`autonomousTraffic`	string	
`bicycle`	string	
`bus`	string	
`delivery`	string	
`emergency`	string	
`motorcycle`	string	
`none`	string	

Name	Type	Introduced
`passengerCar`	string	
`pedestrian`	string	
`simulator`	string	
`taxi`	string	
`throughTraffic`	string	
`truck`	string	
`trucks`	string	

A.3.6 e_road_lanes_laneSection_lr_lane_access_rule

Table 169. Enumerations of the road lanes laneSection lr lane access rule attribute

Name	Type
`allow`	string
`deny`	string

A.3.7 e_laneType

Table 170. Enumerations of the laneType attribute

Name	Type	Introduced	Deprecated	Description
`HOV`	string		1.8.0	High-occupancy vehicle / carpool vehicle. Use `<access>` instead

Name	Type	Introduced	Deprecated	Description
`bidirectional`	string		1.8.0	This lane type has two use cases: a) only driving lane on a narrow road which may be used in both directions; b) continuous two-way left turn lane on multi-lane roads – US road networks Use @direction instead
`biking`	string			Lane that is reserved for cyclists.
`border`	string			Hard border at the edge of the road. It has the same height as the adjacent drivable lane.
`bus`	string		1.8.0	Use `<access>` instead
`connectingRamp`	string			Ramp that connects two motorways, for example, motorway junctions.
`curb`	string	1.6.0		Curb at the edge of the road. Curb stones have a different height than the adjacent drivable lanes.
`driving`	string			Normal drivable road that is not one of the other types.
`entry`	string			Lane that is used for sections that are parallel to the main road and merge into the main road. It is mainly used for acceleration lanes.
`exit`	string			Lane that is used for sections that are parallel to the main road and lead to an exit from the main road. It is mainly used for deceleration lanes.
`median`	string			Lane that sits between driving lanes that lead in opposite directions. It is typically used to separate traffic in towns on large roads.
`mwyEntry`	string		1.5.0	deprecated, use entry instead
`mwyExit`	string		1.5.0	deprecated, use exit instead
`none`	string			Space on the outermost edge of the road. A none lane does not have actual content. Its only purpose is for applications to register that ASAM OpenDRIVE is still present in case the (human) driver leaves the road.

Name	Type	Introduced	Deprecated	Description
`offRamp`	string			Ramp leading away from a motorway and onto rural urban roads.
`onRamp`	string			Ramp leading to a motorway from rural or urban roads.
`parking`	string			Lane with parking spaces.
`rail`	string			Lane used by trains only.
`restricted`	string			Lane on which cars should not drive. The lane has the same height as drivable lanes. Typically, the lane is separated with lines and often contains dotted lines as well.
`roadWorks`	string		1.8.0	
`shared`		1.8.0		Shared by all traffic participants. For shared walking/biking lanes use `<access>`.
`shoulder`	string			Soft border at the edge of the road.
`sidewalk`	string		1.8.0	Use walking instead
`slipLane`	string	1.8.0		On a slip lane a driver can change roads without driving into the main intersection.
`special1`	string		1.8.0	deprecated
`special2`	string		1.8.0	deprecated
`special3`	string		1.8.0	deprecated
`stop`	string			Hard shoulder on motorways for emergency stops.
`taxi`	string		1.8.0	Use `<access>` instead
`tram`	string			Lane used by trams only.

Name	Type	Introduced	Deprecated	Description
`walking`	string	1.8.0		Lane on which pedestrians can walk.

A.3.8 e_road_lanes_laneSection_lcr_lane_roadMark_laneChange

Table 171. Enumerations of the road lanes laneSection lcr lane roadMark laneChange attribute

Name	Type
`both`	string
`decrease`	string
`increase`	string
`none`	string

A.3.9 e_roadMarkColor

Table 172. Enumerations of the roadMarkColor attribute

Name	Type	Introduced	Description
`black`		1.8.0	
`blue`	string		
`green`	string		
`orange`	string		
`red`	string		
`standard`	string		equivalent to "white"

Name	Type	Introduced	Description
`violet`	string		
`white`	string		
`yellow`	string		

A.3.10 e_lane_direction

Table 173. Enumerations of the lane direction attribute

Name	Introduced	Description
`both`	1.8.0	Bidirectional, both directions are valid.
`reversed`	1.8.0	Directly opposite to the standard direction.
`standard`	1.8.0	Direction is determined by the combination of `<left>` or `<right>` lane grouping and the values LHT or RHT of the @rule attribute of a road.

A.3.11 e_roadMarkRule

Table 174. Enumerations of the roadMarkRule attribute

Name	Type
`caution`	string
`no passing`	string
`none`	string

A.4 Object

A.4.1 e_tunnelType

Table 175. Enumerations of the tunnelType attribute

Name	Type	Description
`standard`	string	
`underpass`	string	i.e. sides are open for daylight

A.4.2 e_borderType

Table 176. Enumerations of the borderType attribute

Name	Type	Introduced
`concrete`	string	
`curb`	string	
`paint`	string	1.8.0

A.4.3 e_sideType

Table 177. Enumerations of the sideType attribute

Name	Type
`front`	string
`left`	string

Name	Type
rear	string
right	string

A.4.4 e_outlineFillType

Table 178. Enumerations of the outlineFillType attribute

Name	Type	Introduced
asphalt	string	
cobble	string	
concrete	string	
grass	string	
gravel	string	
paint	string	1.8.0
pavement	string	
soil	string	

A.4.5 e_objectType

Table 179. Enumerations of the objectType attribute

Name	Type	Introduced	Deprecated	Description
barrier	string			A barrier is a continuous roadside object, which cannot be passed.

Name	Type	Introduced	Deprecated	Description
`bike`	string		1.5.0	deprecated
`building`	string			A building is a closed object, which cannot be passed.
`bus`	string		1.5.0	deprecated
`car`	string		1.5.0	deprecated
`crosswalk`	string			A crosswalk is an object on the road that can be passed. It is recommended to be defined as `<crossPath>` within a junction for pedestrian/bicycle simulation. If the crosswalk is defined as an object only, it will not be used for pedestrian/bicycle simulation.
`gantry`	string			A gantry is an object above a road on which `<signals>` are placed.
`motorbike`	string		1.5.0	deprecated
`none`	string			All other objects, that don't fit into existing categories or unknown.
`obstacle`	string			An obstacle is an object on or beside the road that cannot be passed.
`parkingSpace`	string			A parkingSpace is an object on a lane on which vehicles are parked.
`patch`	string		1.8.0	use roadSurface instead
`pedestrian`	string		1.5.0	deprecated
`pole`	string			A pole is a thin long object.
`railing`	string		1.8.0	use barrier instead
`roadMark`	string			A roadMark object is painted on the road and can be passed.
`roadSurface`	string	1.8.0		A roadSurface object is on the road and can be passed.

Name	Type	Introduced	Deprecated	Description
`soundBarrier`	string		1.8.0	use barrier instead
`streetLamp`	string		1.8.0	use pole instead
`trafficIsland`	string			A trafficIsland object is on the road and should not be passed by vehicles.
`trailer`	string		1.5.0	deprecated
`train`	string		1.5.0	deprecated
`tram`	string		1.5.0	deprecated
`tree`	string			A tree object is a single vegetational object with a trunk.
`van`	string			
`vegetation`	string			A vegetation object is a single vegetational object without a trunk or an area of vegetation.
`wind`	string		1.5.0	deprecated, use pole instead

A.4.6 e_bridgeType

Table 180. Enumerations of the bridgeType attribute

Name	Type
`brick`	string
`concrete`	string
`steel`	string
`wood`	string

A.4.7 e_orientation

Table 181. Enumerations of the orientation attribute

Name	Type
`+`	string
`-`	string
`none`	string

A.4.8 e_road_objects_object_parkingSpace_access

Table 182. Enumerations of the road objects object parkingSpace access attribute

Name	Type
`all`	string
`bus`	string
`car`	string
`electric`	string
`handicapped`	string
`residents`	string
`truck`	string
`women`	string

A.5 Railroad

A.5.1 e_station_type

Table 183. Enumerations of the station type attribute

Name	Type
`large`	string
`medium`	string
`small`	string

A.5.2 e_road_railroad_switch_position

Table 184. Enumerations of the road railroad switch position attribute

Name	Type
`dynamic`	string
`straight`	string
`turn`	string

A.5.3 e_station_platform_segment_side

Table 185. Enumerations of the station platform segment side attribute

Name	Type
`left`	string
`right`	string

A.6 Road

A.6.1 e_road_link_elementType

Table 186. Enumerations of the road link elementType attribute

Name	Type
`junction`	string
`road`	string

A.6.2 e_paramPoly3_pRange

Table 187. Enumerations of the paramPoly3 pRange attribute

Name	Type
`arcLength`	string
`normalized`	string

A.6.3 e_roadType

Table 188. Enumerations of the roadType attribute

Name	Type
`bicycle`	string
`lowSpeed`	string
`motorway`	string

Name	Type
pedestrian	string
rural	string
townArterial	string
townCollector	string
townExpressway	string
townLocal	string
townPlayStreet	string
townPrivate	string
town	string
unknown	string

A.6.4 e_strip_mode

Table 189. Enumerations of the strip mode attribute

Name	Description
independent	height values due to cross section surfaces are calculated independent of the inner strip
relative	height values due to cross section surfaces are added to the height values of the outer edge of the inner strip

A.6.5 e_maxSpeedString

Table 190. Enumerations of the maxSpeedString attribute

Name	Type
`no limit`	string
`undefined`	string

A.6.6 e_trafficRule

Table 191. Enumerations of the trafficRule attribute

Name	Type
`LHT`	string
`RHT`	string

A.6.7 e_direction

Table 192. Enumerations of the direction attribute

Name	Type
`opposite`	string
`same`	string

A.6.8 e_countryCode_deprecated

Table 193. Enumerations of the countryCode deprecated attribute

Name	Type
`Austria`	string

Name	Type
`Brazil`	string
`China`	string
`France`	string
`Germany`	string
`Italy`	string
`OpenDRIVE`	string
`Switzerland`	string
`USA`	string

A.7 Signal

A.7.1 e_road_signals_displayType

Table 194. Enumerations of the road signals displayType attribute

Name	Introduced	Description
`LED`	1.8.0	Full LED boards
`monochromGraphic`	1.8.0	Yellow or white text as lights on black background
`other`	1.8.0	All other display types that do not fit into current categories
`rotatingPrismHorizontal`	1.8.0	No lights. Horizontal rotating metal prism.
`rotatingPrismVertical`	1.8.0	No lights. Vertical rotating metal prism.

Name	Introduced	Description
`simpleMatrix`	1.8.0	Outside is fixed, content inside changes.

A.7.2 e_signals_semantics_supplementaryTime

Table 195. Enumerations of the signals semantics supplementaryTime attribute

Name	Introduced
`day`	1.8.0
`time`	1.8.0

A.7.3 e_signals_semantics_priority

Table 196. Enumerations of the signals semantics priority attribute

Name	Introduced
`4way`	1.8.0
`keepClearLine`	1.8.0
`noParkingLine`	1.8.0
`noTurnOnRed`	1.8.0
`priorityRoadEnd`	1.8.0
`priorityRoad`	1.8.0
`priorityToTheRightRule`	1.8.0
`stopLine`	1.8.0

Name	Introduced
`stop`	1.8.0
`trafficLight`	1.8.0
`turnOnRedAllowed`	1.8.0
`waitingLine`	1.8.0
`yield`	1.8.0

A.7.4 e_road_signals_signal_reference_elementType

Table 197. Enumerations of the road signals signal reference elementType attribute

Name	Type
`object`	string
`signal`	string

A.7.5 e_signals_semantics_lane

Table 198. Enumerations of the signals semantics lane attribute

Name	Introduced
`noOvertakeCarsEnd`	1.8.0
`noOvertakeCars`	1.8.0
`noOvertakeTrucksEnd`	1.8.0
`noOvertakeTrucks`	1.8.0

Name	Introduced
`priorityOverOncoming`	1.8.0
`roundabout`	1.8.0
`yieldForOncoming`	1.8.0

A.7.6 e_signals_semantics_supplementaryEnvironment

Table 199. Enumerations of the signals semantics supplementaryEnvironment attribute

Name	Introduced
`fog`	1.8.0
`rain`	1.8.0
`snow`	1.8.0

A.7.7 e_signals_semantics_speed

Table 200. Enumerations of the signals semantics speed attribute

Name	Type	Introduced
`maximumEnd`	string	1.8.0
`maximum`	string	1.8.0
`minimumEnd`	string	1.8.0
`minimum`	string	1.8.0
`recommendedEnd`	string	1.8.0

Name	Type	Introduced
`recommended`	string	1.8.0
`zoneEnd`	string	1.8.0
`zone`	string	1.8.0

A.7.8 e_signals_semantics_supplementaryDistance

Table 201. Enumerations of the signals semantics supplementaryDistance attribute

Name	Introduced
`for`	1.8.0
`in`	1.8.0